Customer Segmentation and Data-driven Sales Strategy for Turtle Games - Technical Report

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Background: Turtle Games is a game manufacturer and retailer and they would like to improve overall sales performance by understanding and utilising customer trends.

Turtle Games need a data-informed approach to customer segmentation, prioritisation, sentiment analysis and to using loyalty points data for predicting and informing sales performance and the findings aim to support targeted interventions to understand and improve sales

This analysis focusses on understanding customer demographics and spending behaviour along with loyalty point data to develop predictive models that can inform sales strategy.

Business questions:

1. What factors influence loyalty point accumulation?
2. Can customers be grouped into meaningful segments based on age, income (remuneration), or spending habits?
3. How do customers feel about products and experiences based on their reviews and feedback?
4. Can we leverage the loyalty points data to develop descriptive statistics and provide insights into where should the business focus its efforts for maximum impact?

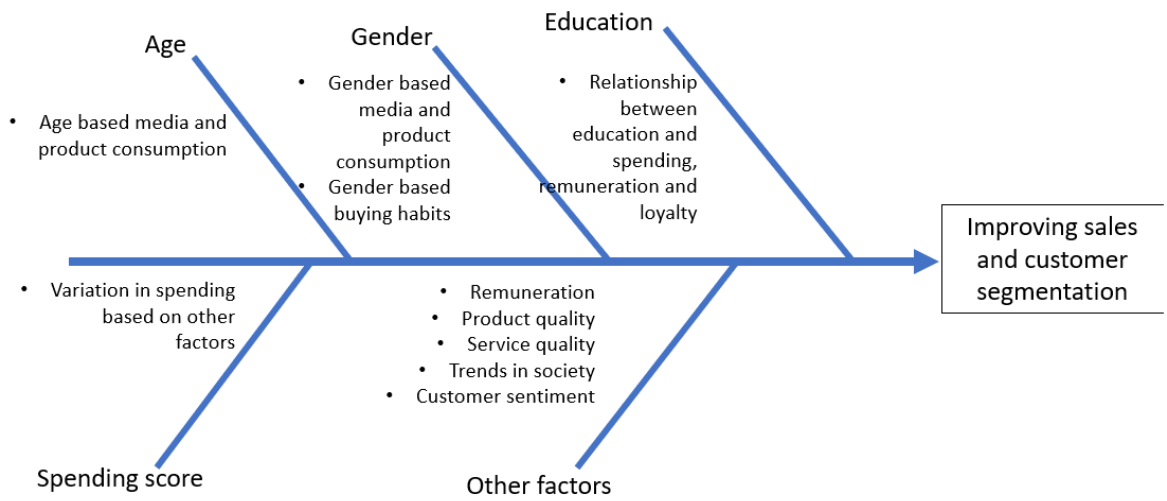
Analytical questions:

The data is approached with the following questions to address the business questions:

1. Assess current loyalty point accumulation and it's relationship with customer age, remuneration, education, gender and spending habits.
2. Identify the parameters that have the strongest impact on loyalty point accumulation to drive customer segmentation.
3. Analysing customer reviews and feedback to assess customer sentiment and it's variance across segments.
4. Predicting loyalty point accumulation based on relevant customer data to inform marketing approach.

Analytical approach:

The exploratory analysis of this dataset has been approached through the **5-Whys framework**. The fishbone diagram of various factors is as given below:



Data Source:

Turtle Games has provided data in the following files:

- turtle_reviews.csv – Details on customer gender, age, remuneration, spending score, loyalty points, education, language, platform, review and summary across products.
- metadata_turtle_games.txt – Details of the data set, data quality, and reference.

Libraries:

The libraries used for this analysis are as below:

1. Linear regression using Python

```
# Import libraries
import numpy as np
import pandas as pd
import statsmodels.api as sm
import statsmodels.stats.api as sms
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from statsmodels.stats.outliers_influence import variance_inflation_factor
from sklearn import metrics

import warnings
warnings.filterwarnings('ignore')
```

2. Exploring the structure using decision trees

```python
# Import all the necessary packages
from sklearn.preprocessing import LabelEncoder
from sklearn.tree import DecisionTreeRegressor
from sklearn.preprocessing import OneHotEncoder
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score
from sklearn.tree import plot_tree
```

3. Clustering with k-means using Python

```python
# Import necessary libraries.
import matplotlib.cm as cm

from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score

import warnings
warnings.filterwarnings('ignore')
```

4. NLP using Python

```python
# Import all the necessary packages.

import nltk
import os
import re
import contractions


nltk.download('punkt')
nltk.download('stopwords')

from wordcloud import WordCloud
from nltk.tokenize import word_tokenize
from nltk.probability import FreqDist
from nltk.corpus import stopwords
from textblob import TextBlob
from scipy.stats import norm

from nltk.sentiment.vader import SentimentIntensityAnalyzer

from sklearn.linear_model import LogisticRegression
from sklearn.metrics import classification_report, confusion_matrix

# Import Counter.
from collections import Counter

import warnings
warnings.filterwarnings('ignore')
```

5. Load, clean, wrangle and analyse data using R

```r
# Load libraries
library(tidyverse)  # Data manipulation & visualisation
library(corrplot)   # Correlation plots
library(moments)    # Skewness & kurtosis
```

Functions:

Considering repeated usage of certain actions, the following functions were defined:

1. plot_regression to create a scatter plot with regression line
2. preprocess_text to expand contractions, remove URLs, hashtags, stop words and special characters, convert to lowercase
3. sentiment_label (positive, negative, neutral)
4. top_words_by_cluster
5. generate_polarity

Ingestion & schema validation:

The .csv datasets were imported into Pandas DataFrames for analysis. Initial validation steps included verifying record counts, expected column names, and data types to ensure schema

integrity. Data fields were coerced to appropriate formats; for example, converting objects to str to prevent computational errors and enable sentiment analysis.

Standardisation & cleaning in Python:

Data was checked for duplicate rows and missing values, none were found.

Record counts and date ranges of the datasets were determined to ensure data is within expected range.

A small number of records exhibit implausible combinations of age and highest educational qualification (e.g. PhD holders under the age of 23 or graduate at age 18). These observations were retained as they do not materially affect the regression results but are noted as a data quality limitation.

education	count	mean	std	min	25%	50%	75%	max
Basic	50.0	42.000000	7.532244	32.0	34.0	46.0	49.0	49.0
PhD	460.0	40.304348	14.921537	17.0	32.0	38.0	45.0	72.0
diploma	190.0	47.842105	11.975562	32.0	37.0	46.0	56.0	70.0
graduate	900.0	36.855556	12.520560	18.0	28.0	36.0	45.0	70.0
postgraduate	400.0	40.225000	13.703790	17.0	29.0	37.0	50.0	72.0

	age	education	loyalty_points
0	18	graduate	210
5	24	PhD	573
17	22	PhD	579
33	17	PhD	1042
35	23	PhD	1152

Renamed columns "remuneration (k£)": "remuneration", "spending_score (1-100)": "spending_score" for ease of analysis.

Dropped unnecessary columns: language (All English) and platform (all Web) as it added no value to the analysis.

Saved csv file as 'turtle_reviews_clean.csv' which was used for further analysis, including in R.

NLP: Change to lower case and join the elements in each of the columns respectively Replace punctuation in each of the columns respectively, cleaned versions of the test using preprocess_text function.

Cleaning and validation Steps in R:

The cleaned csv was imported in R and data is validated by checking summary, missing values and data types to ensure correct import.

Feature engineering:

For using Decision Tree: "Review" and "Summary" columns are dropped as they have unstructured text data which is not useful for this analysis. "Product" has product ID which is

also not useful for this analysis and is hence dropped. Used one-hot encoding to encode categorical columns and changed boolean column to binary so that trees can handle them. Checked missing values after conversion to ensure no errors.

For K-means clustering: Dropped unnecessary columns- 'gender', 'age', 'remuneration', 'spending_score', 'loyalty_points', 'education'

For NLP: Dropped unnecessary columns- 'gender', 'age', 'remuneration', 'spending_score', 'loyalty_points', 'education', 'product']. Added cluster labels into the NLP dataframe.

	cluster_name	review_clean
0	Budget Essentials	comes dms screen space screen absolute premium...
1	Aspiring Spenders	open letter galeforce9 unpainted miniatures no...
2	Budget Essentials	nice art nice printing two panels filled gener...
3	Aspiring Spenders	amazing buy bought gift new dm perfect
4	Budget Essentials	review gf9s previous screens completely unnece...

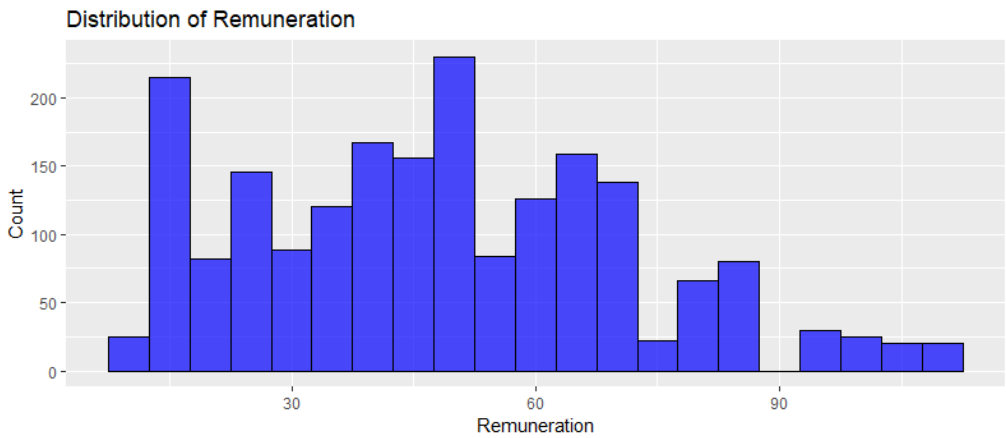
Exploratory aggregation & visuals:

Exploratory aggregation and visualisation examined trends and correlation between various features and loyalty points. Boxplots identified outliers; histogram showed distribution, bar charts compared clusters and scatter plots showed correlation. Word clouds and sentiment scores provided qualitative context to customer attitudes. Used groupby for aggregating data.

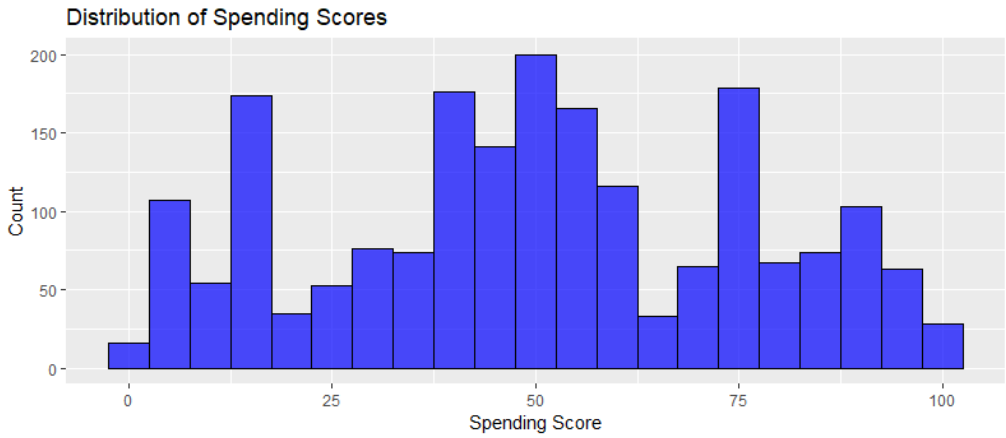
Trends and patterns from Exploratory analysis:

1. **Distribution of remuneration:**

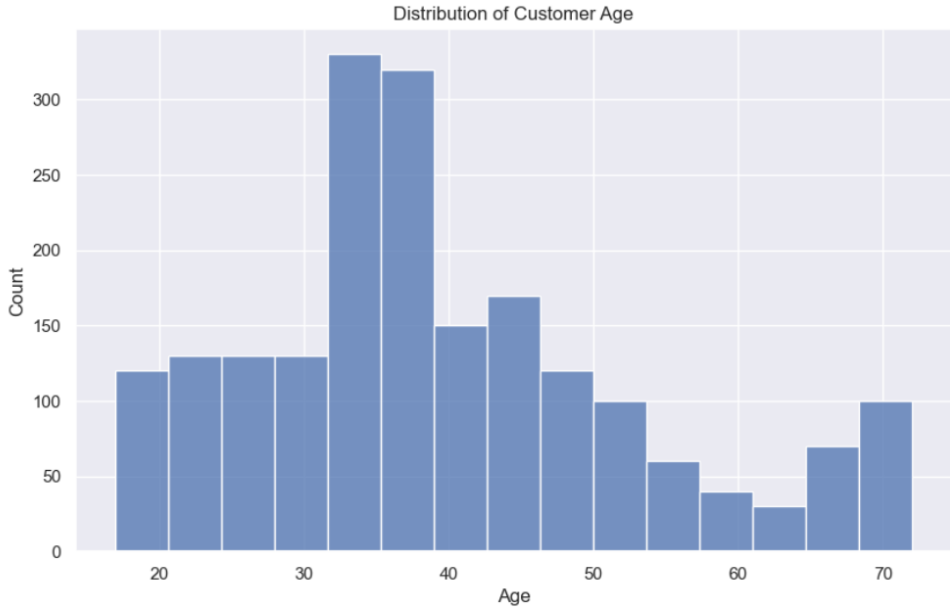
   Count of remuneration suggests a mild right skew and a smaller number of high-income customers.



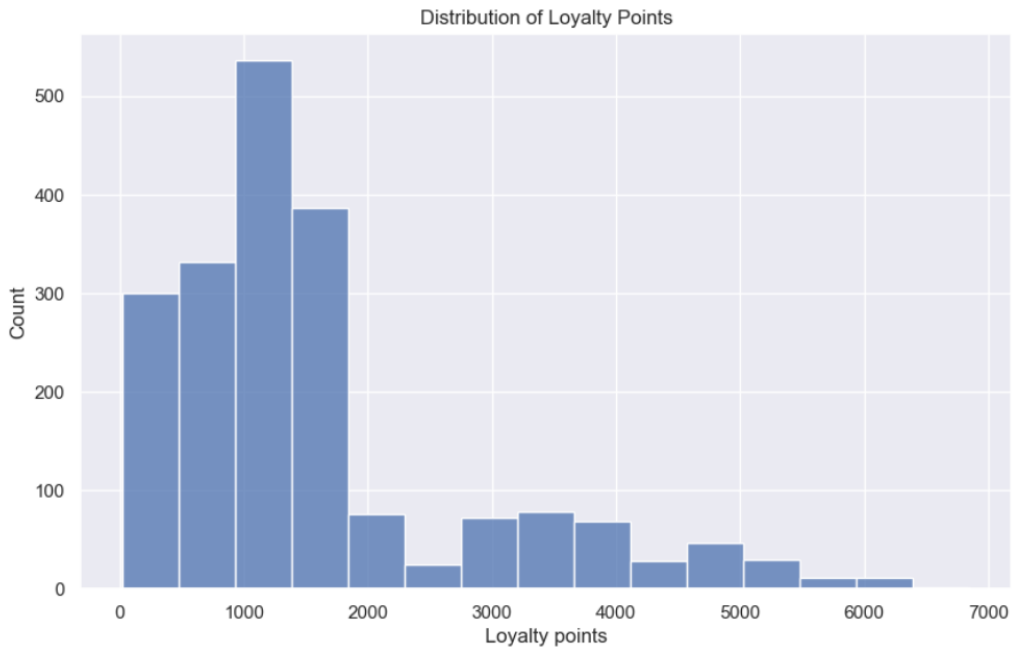
2. **Distribution of spending scores**
   Spending scores are evenly distributed, indicating a balanced customer base.



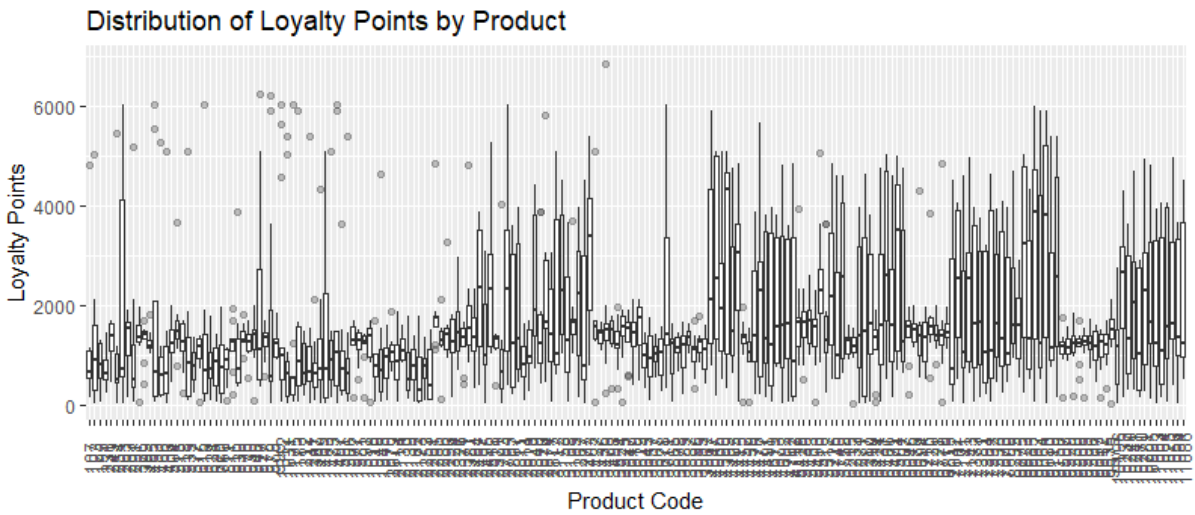
3. **Distribution of age**
   Count of customer age is dominantly around middle adulthood (30-40 years), consistent with a working-age customer base.



4. **Distribution of loyalty points**
   Loyalty points show a wide spread.



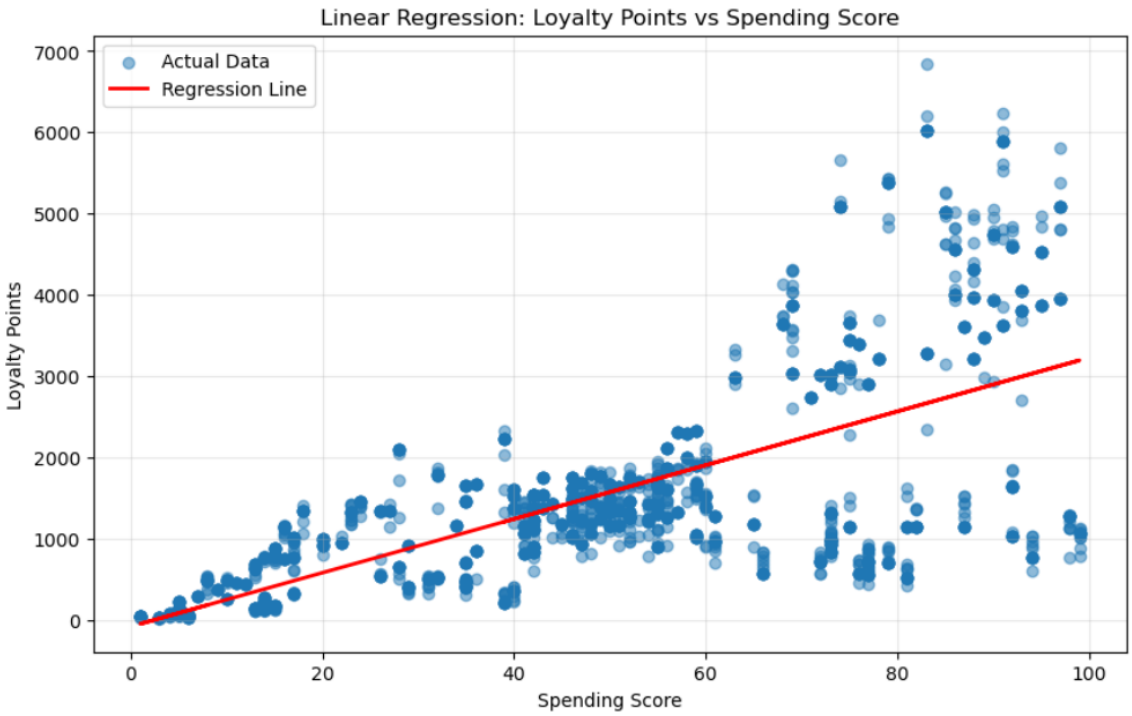
5. **Distribution of loyalty points with products**
   Some products generate substantially more loyalty points than others.
   On average, products with a higher spending score are more positively related to higher average loyalty points.



Regression analysis

Three models were built using spending score, remuneration and age. The scatter plots and r2 values are as below.

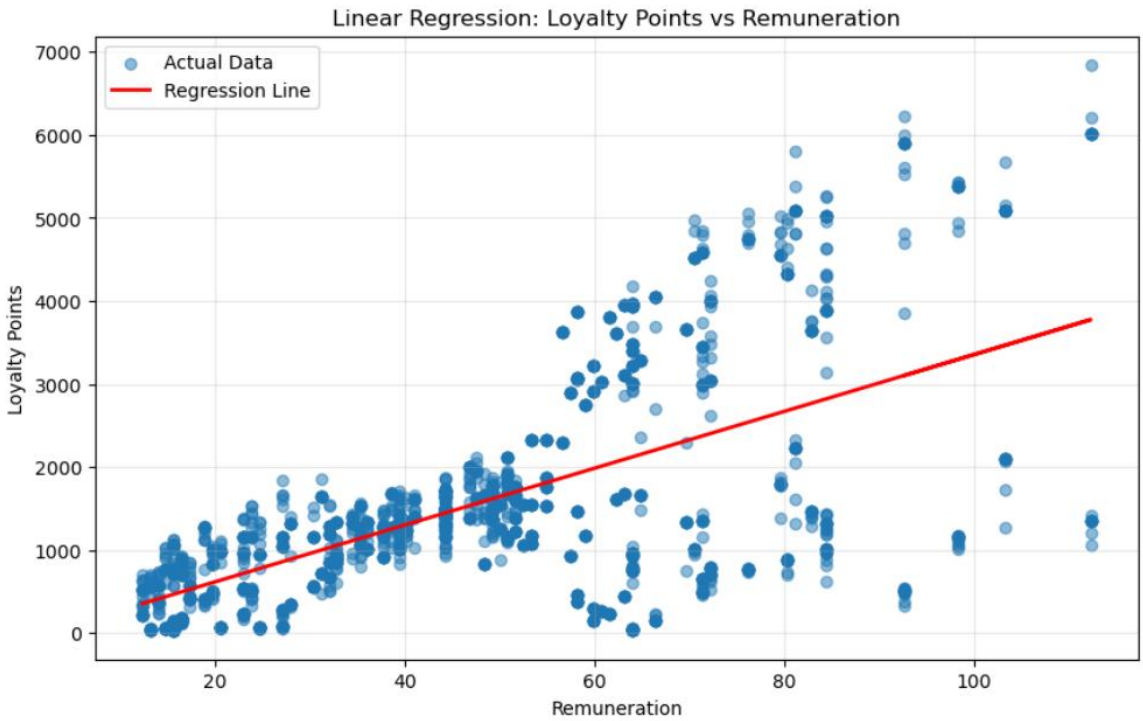
Model 1: Spending score




```
Regression Equation:
Loyalty Points = -75.0527 + 33.0617 * Spending Score
R-squared: 0.4520
```

p-value ($p < 0.001$) confirms that this relationship is significant.

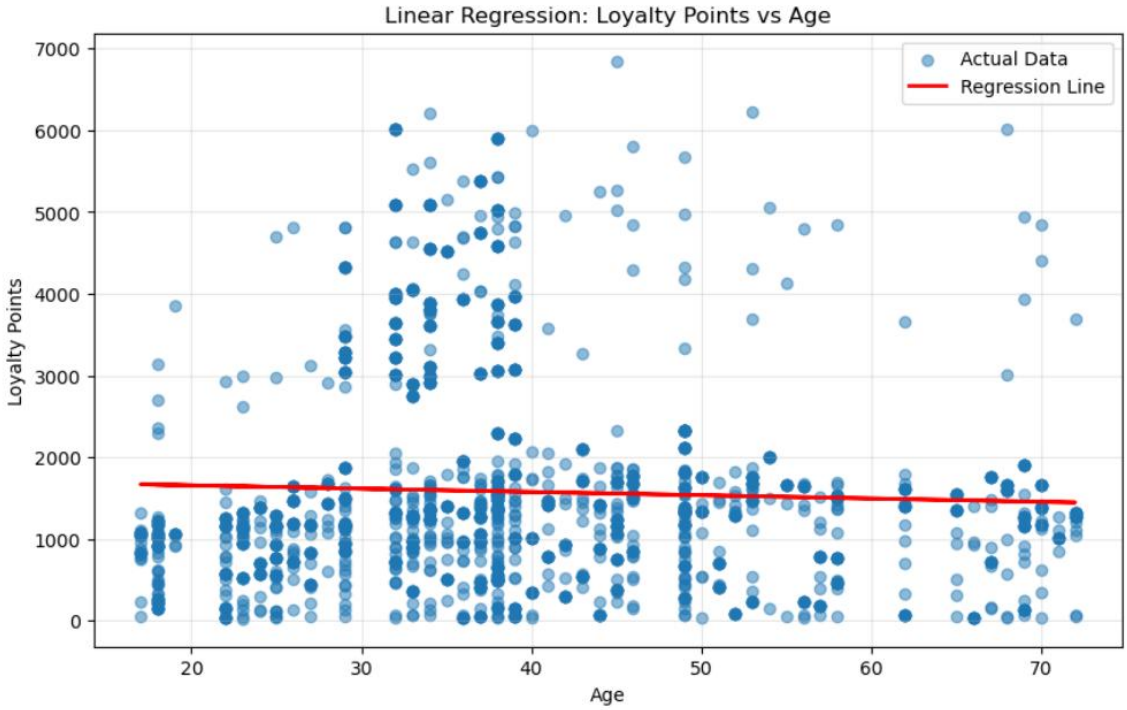
Model 2: Remuneration




```
Regression Equation:
Loyalty Points = -65.6865 + 34.1878 * Remuneration
R-squared: 0.3795
```

The p-value ($p < 0.001$) confirms that this relationship is significant.

Model 3: Age




```
Regression Equation:
Loyalty Points = 1736.5177 + -4.0128 * Age
R-squared: 0.0018
```

No clear relationship is seen between age and loyalty points ($p = 0.058$). Age does not appear to be a useful predictor of loyalty point accumulation.

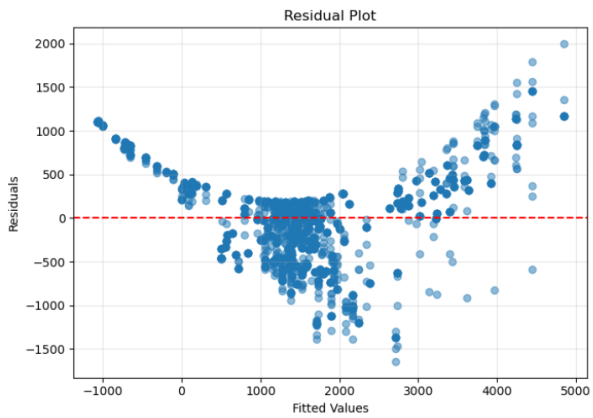
Multiple Linear Regression:

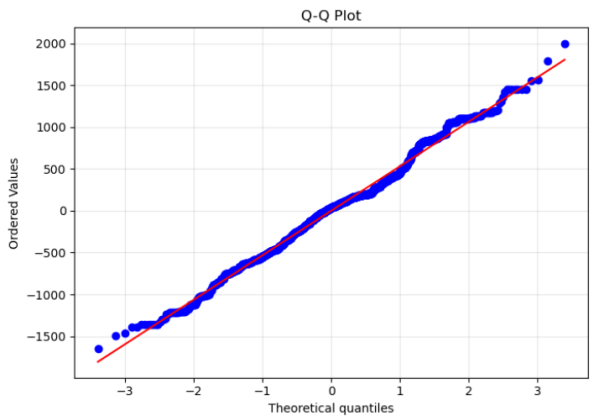
Model is built based on spending_score and remuneration and age is excluded due to insignificance.

```
                            OLS Regression Results
==============================================================================
Dep. Variable:         loyalty_points   R-squared:                       0.827
Model:                            OLS   Adj. R-squared:                  0.827
Method:                 Least Squares   F-statistic:                     4770.
Date:                Mon, 12 Jan 2026   Prob (F-statistic):               0.00
Time:                        10:17:14   Log-Likelihood:                -15398.
No. Observations:                2000   AIC:                         3.080e+04
Df Residuals:                    1997   BIC:                         3.082e+04
Df Model:                           2
Covariance Type:            nonrobust
==================================================================================
                     coef    std err          t      P>|t|      [0.025      0.975]
----------------------------------------------------------------------------------
const          -1700.3051     35.740    -47.575      0.000   -1770.396   -1630.214
spending_score    32.8927      0.458     71.845      0.000      31.995      33.791
remuneration      33.9795      0.517     65.769      0.000      32.966      34.993
==============================================================================
Omnibus:                        4.723   Durbin-Watson:                   3.477
Prob(Omnibus):                  0.094   Jarque-Bera (JB):                4.650
Skew:                           0.103   Prob(JB):                       0.0978
Kurtosis:                       3.115   Cond. No.                         220.
==============================================================================
```

Both predictors and F-statistic are highly significant, indicating suitability of this model to predict loyalty points based on spending behaviour and remuneration.

Residuals, VIF, accuracy (MAE, MSE, RMSE) were checked and found to be within acceptable range, allowing the model to be used to predict business case scenarios.






```
--- Variance Inflation Factor (VIF) ---
         Variable       VIF
0  spending_score  2.779416
1    remuneration  2.779416
```

Given the strong model fit, low multicollinearity (<5), and large sample size, the estimated coefficients remain reliable and the model is appropriate for inference and prediction.

```
--- Prediction Scenarios ---
   spending_score  remuneration  predicted_loyalty_points
0              25            25                 -28.500259
1              50            50                1643.304578
2              75            75                3315.109416
3              90           100                4657.987307
```

These predictions can help Turtle Games estimate loyalty points for new customers.

```
---MLR Model Accuracy Metrics ---
Mean Squared Error (MSE): 284879.79
Root Mean Squared Error (RMSE): 533.74
Mean Absolute Error (MAE): 414.83
```

Analysis in R:

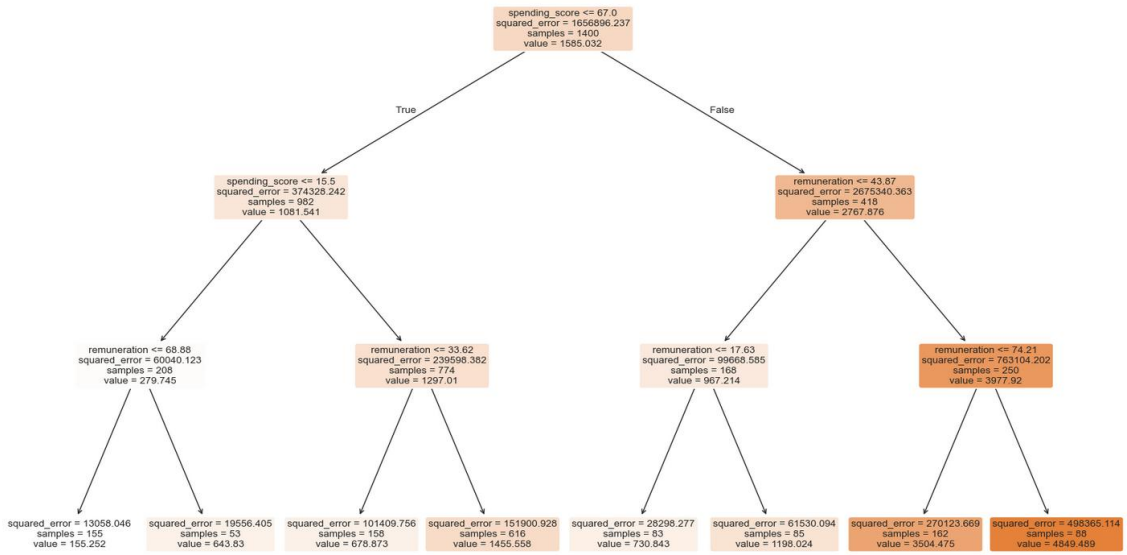
Two models for MLR were built in R.

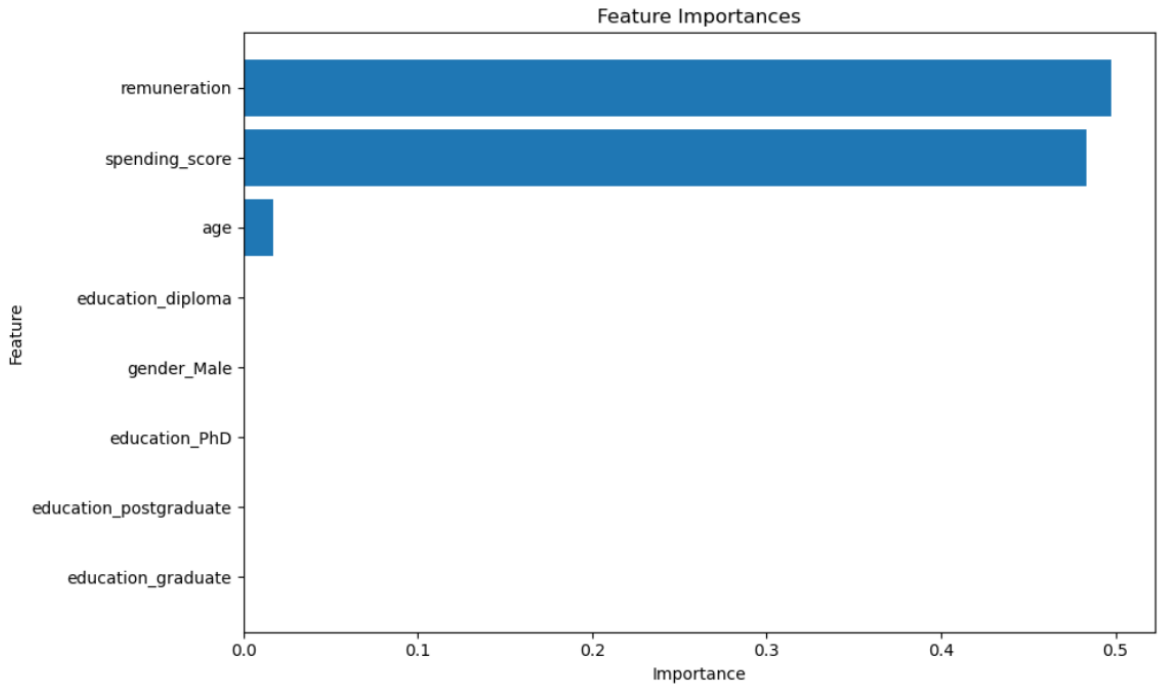
Model 1: Same as python

Model 2: Age added along with remuneration and spending score

Age is not independently significantly correlated with loyalty points but is seen to be statistically significant in this MLR model. However, adjusted R-squared increases only by ~1.3 percentage points. To avoid complexity, the final model retained only spending score and remuneration as predictors.

Decision Trees:

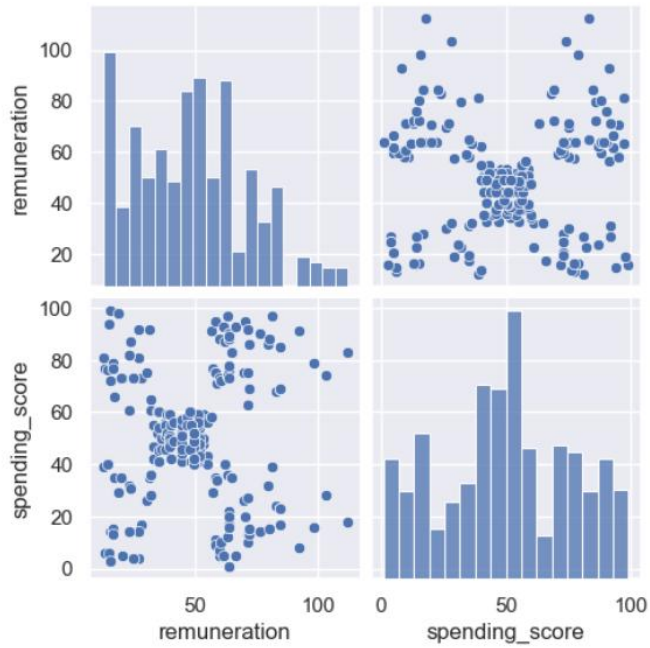




Spending score and remuneration are seen to be the dominant features affecting loyalty points, followed by age. Education indicators and Gender have a minor to no impact.

After pruning the tree, despite the RMSE increasing, the model retains strong explanatory power ($R^2 \approx 0.91$), indicating that the key drivers of loyalty points were preserved.

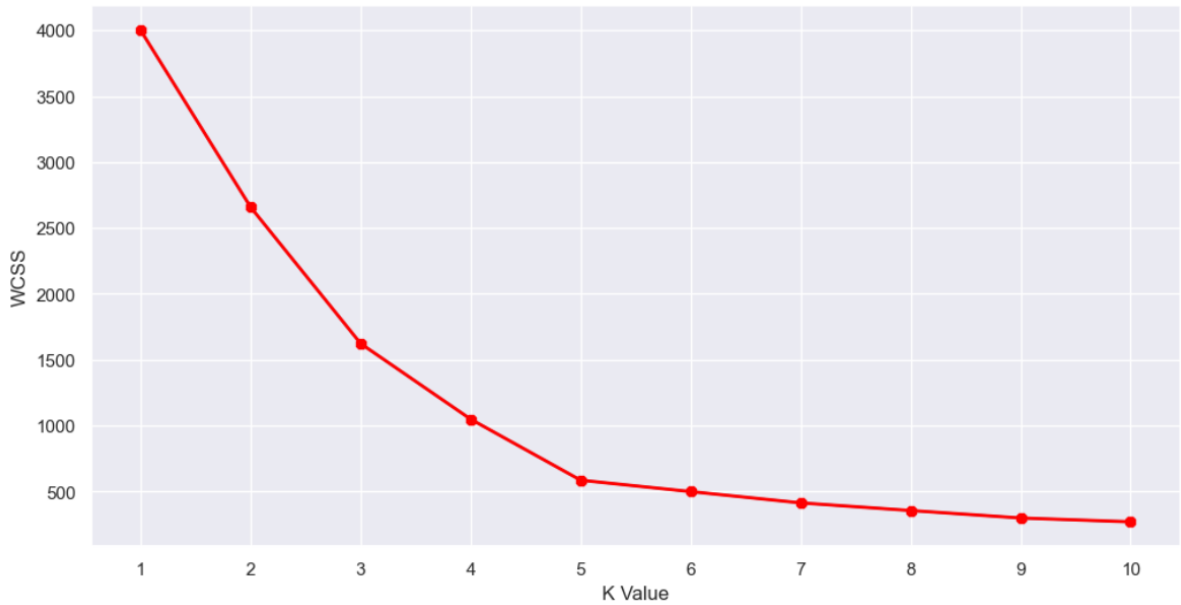
K-means clustering:

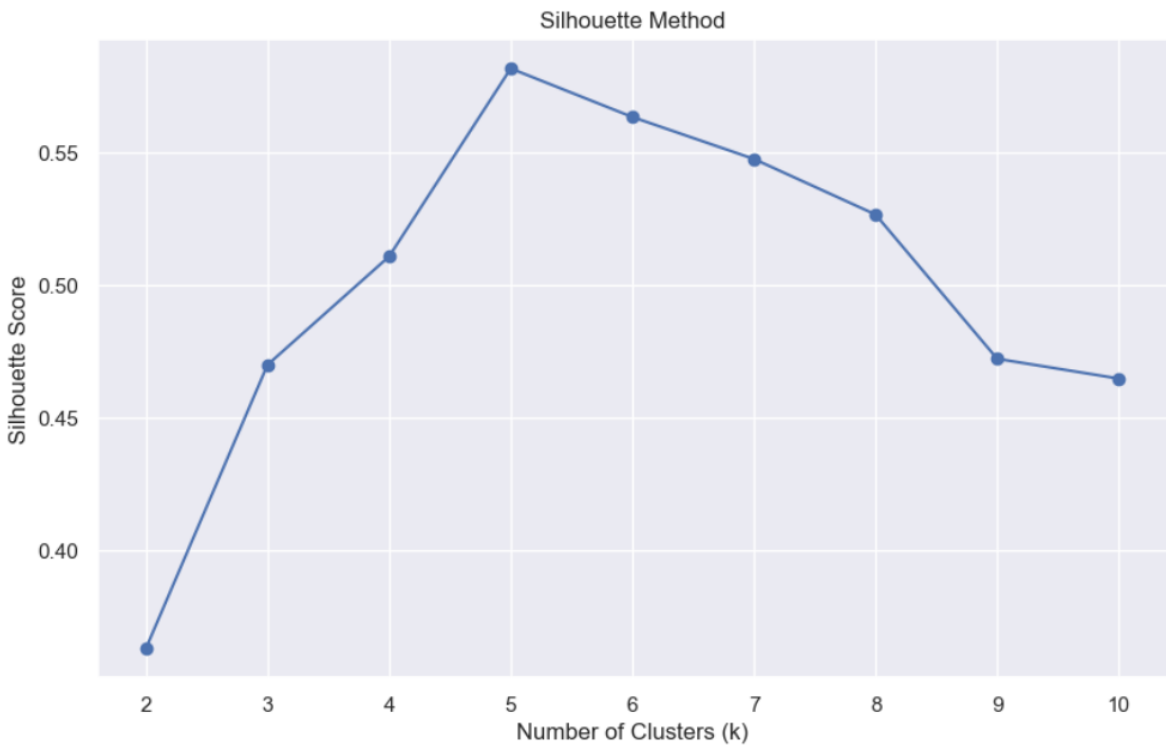


Visually, some level of clustering is observed between remuneration and spending score.

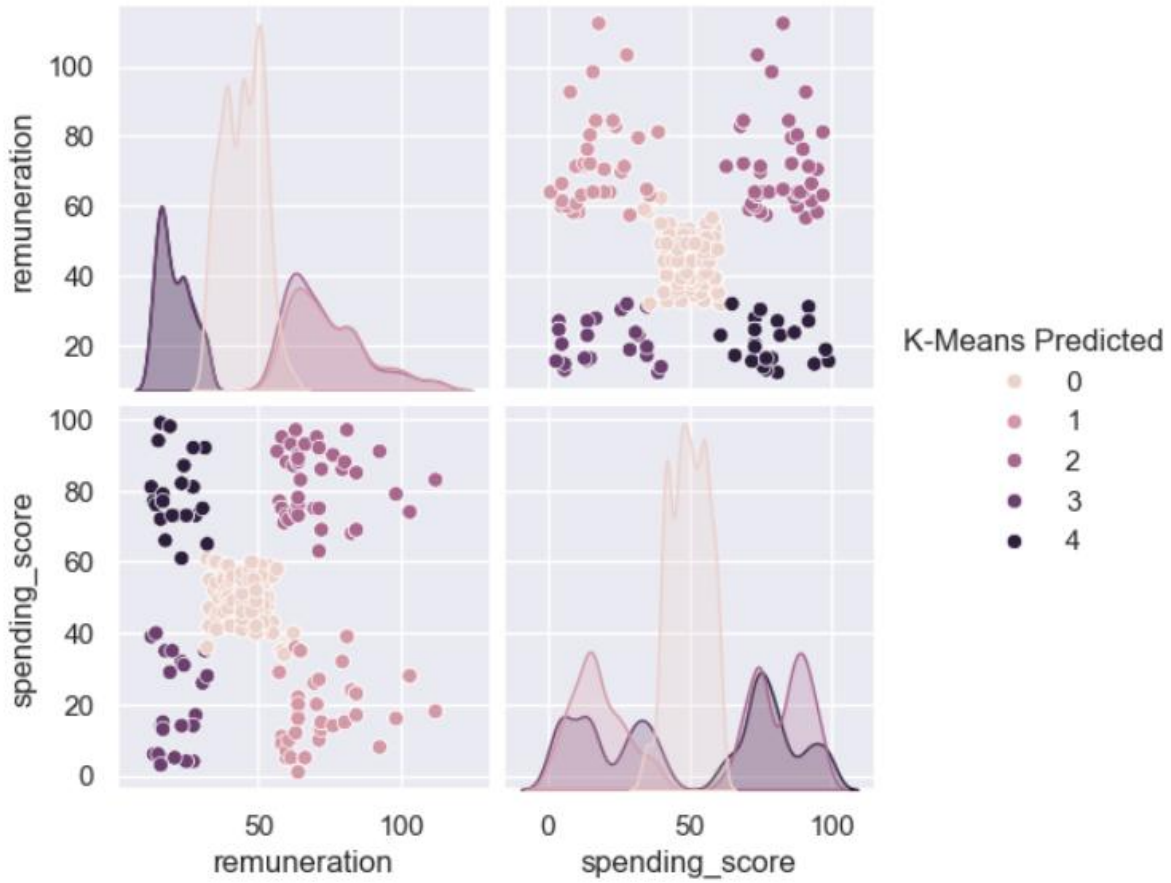
Standardisation is applied on the dataset to allow clustering.

The Elbow and Silhouette methods are used to identify an optimal number of clusters by balancing compactness and separation.





The elbow and peak score are at k = 5. Accordingly, 4, 5 and 6 clusters can be used for evaluating the model.



k = 4 clusters leads to under-segmentation, whereas, k = 6 leads to over-segmentation. Therefore, k = 5 was selected as it provides an appropriate balance between all clusters.

Five distinct customer types observed from k-means clustering. These clusters are labelled based on their income and spending patterns.

Five clusters are identified based on remuneration and spending score. The recommended actions and customer profiles are as tabulated below.

Cluster Name	Remuneration	Spending Behaviour	Customer Profile	Business Opportunity	Recommended Actions
Budget Essentials	Low	Low	Cautious, necessity-driven customers with potentially high price sensitivity	Stable but low-margin segment	Promotions on essentials, discount bundles, value packs
Aspiring Spenders	Low–Medium	High	Customers spending disproportionately relative to income	High growth and loyalty potential	Rewards programs, BNPL options, targeted promotional offers
Balanced Shoppers	Medium	Medium	Core customer base with predictable, value-oriented purchasing behaviour	Reliable revenue contributors	Retention campaigns, cross-selling, personalised recommendations
Premium Enthusiasts	High	High	High-value customers with strong purchasing power and loyalty potential	Key profit-driving segment	VIP programs, exclusive offers, early product access
Affluent Conservatives	High	Low	Financially capable customers with conservative spending habits	Untapped revenue opportunity	Product education, premium trials, targeted nudges

NLP analysis

NLP was used to generate wordclouds from review and summary columns, polarity scores using textblob and VADER sentiment analysis and top 20 positive and negative words, along with cluster based sentiment analysis.

game
great
play
one
fun
time
card
love
make
player
well
friend
expansion
little
playing
book
played
take
piece
good
many
even
find
first
rule
use
much
work
people
version
think
bought
add
thing
box
point
shape
feel
need
got
together
using
dungeon
monster
loved
quality
puzzle
nice
used
hard
another
product
hour
toy
way
enjoy
egg
buy
help
set
different
new
really
still
board
game
dd
kid
robot
gift
always
great game
small
children
seem
three
recommend
give
made
go
turn
come
cute
want
sticker
better
making
say
found
looking
already
bit
difficult
group
review
know
yarn
age
easy
fit
strategy
family
lot
enough
two
may
thought
monster
loved
best
instruction
awesome
gets
able
try
day
board
hand
probably
item
place
stock
every
must
mini
hold
adult
others
game play
far
draw
quest
fan
quite
waterdeep
lords
price
definitely
perfect
look
animal
daughter
magnet
acquire
building
plastic
never
letter
kit
read
original
keep
adventure
us
start
money
long
back
picture
overall
character
love game
size
right
question
pretty
anger
everyone
open
part
without
purchased
cardboard
actually
see
ball whacks
learn
fun
getting
around
though
child
interesting
everything
excellent
number
looking
found
better
making
strategy
family
base game
yarn
difficult
review
group
bit
know
easily
favorite
go
ball
color
three
might
something
end
big
picture
back
long
money
start
adventure
us
read
kit
original
keep
never
letter
plastic
great game
small
children
seem
three
recommend
give
made
go
turn
come
cute
want
sticker
better
making
say
found
looking
already
bit
difficult
group
review
know
yarn
age
easy
fit
strategy
family
lot
people
version
think
bought
awesome
gets
able
try
day
board
hand
probably
item
place
stock
every
must
mini
hold
best
instruction
box
point
shape
feel
need
got
together
using
dungeon
monster
loved
quality
puzzle
nice
used
hard
another
product
hour
toy
way
enjoy
egg
buy
help
set
different
new
really
still
board
game
dd
kid
robot
gift
always

Both reviews and summary indicate a neutral to generally positive polarity. Sentiment analysis revealed similar patterns in cluster-based analysis.

```
[('stars', 427),
 ('five', 342),
 ('game', 319),
 ('great', 295),
 ('fun', 218),
 ('love', 93),
 ('good', 92),
 ('four', 58),
 ('like', 54),
 ('expansion', 52),
 ('kids', 50),
 ('cute', 45),
 ('book', 43),
 ('one', 38),
 ('awesome', 36)]
```

The top 15 most frequently occurring words are quite positive and there is no mention of any concerns such as price, quality or service. This reflects overall customer satisfaction.

Conclusions and next steps:

Conclusions and next steps

- Turtle Games' customer base is diverse.
- Spending behaviour along with remuneration are the strongest drivers of loyalty.
- Customer segmentation along spending score and remuneration enables smarter targeting, better retention, and higher lifetime value.
- Overall sentiment is generally positive
- Loyalty point accumulation and customer segmentation can together be used for defining sales and marketing strategies using predictive models.

Next steps:

- Repeat the analysis in 6 months time following implementation of interventions
- Include more features in data collection such as product categories, customer tenure, basket sizes and customer ID for frequency analysis

Business recommendations:

Business recommendations

1. Prioritise High-Value Customer Segments with Tailored Engagement

(High Priority | Short-term: 0–6 months)

Why:
Clustering analysis identified a distinct **"Premium Enthusiasts"** segment with both high remuneration and high spending. Retaining and deepening engagement with this group offers the fastest return on investment.

Actions:
- Introduce VIP loyalty tiers with exclusive rewards and early product access.
- Personalise marketing campaigns using spending and sentiment signals.
- Offer premium bundles or limited-edition products to reinforce exclusivity.

KPIs:
- 5–10% increase in average loyalty points per Premium customer within 6 months.
- Improvement in repeat purchase rate for this segment.
- Positive uplift in sentiment scores for premium-targeted communications.

Business recommendations

2. Activate Underperforming but High-Potential Customers

(Medium Priority | Medium-term: 6–12 months)

Why:
The **"Affluent Conservatives"** show high remuneration but relatively low spending, indicating untapped revenue potential.

Actions:
- Deploy targeted education and trial campaigns to reduce perceived purchase risk.
- Use personalised incentives (e.g. first-purchase bonuses, premium trials).
- Highlight product value and long-term benefits in communications.

KPIs:
- Increase average spending score of this cluster by 10% within 12 months.
- Growth in total loyalty points generated by this segment.
- Conversion rate from low to medium spending behaviour.

Business recommendations

3. Protect and Retain the Core Customer Base

(High Priority | Ongoing)

Why:
"Balanced Shoppers" form the largest proportion of the customer base and provides stable, predictable revenue. While individually less valuable than premium customers, the business's long-term performance depends heavily on keeping this group engaged.

Actions:

- Maintain consistent loyalty rewards and retention-focused campaigns.
- Cross-sell complementary products based on past purchasing patterns.
- Monitor sentiment trends to proactively address dissatisfaction.

KPIs:

- Stable or increasing retention rate within the Balanced Shoppers segment.
- Sustained contribution to total loyalty points.
- Reduction in negative sentiment signals within this cluster.